A2 · Unit 4 · WPH14 | Further Mechanics, Fields & Particles

Electric Fields

Rub a balloon on your jumper and it can lift your hair without touching it: a charged object fills the space around it with an **electric field**. In this topic we define field strength, use Coulomb's law for the force between two charges, meet electric potential, and see how field and potential describe the same thing in two different ways. These ideas run straight on into capacitors, particle accelerators and the deflection of electron beams.

SPEC 92 Electric fields as force fields

WHAT THIS SPEC POINT ASKS

An electric field is a region where a charged particle experiences a force.

- Every charge sets up an electric field around itself. Any **other** charge placed in that field feels a force.
- **Like** charges repel; **unlike** charges attract.
- The **direction of an electric field** at a point is the direction of the force on a small **positive** test charge placed there. A negative charge feels a force the opposite way.
- So field lines point **away from positive** charges and **towards negative** charges.
- Electric forces act on **charges whether they move or not**. Compare: a gravitational field acts on **mass**, and a magnetic field acts on **moving** charges (and currents).

Field	Acts on	Defined as	Unit
Gravitational	mass	force per unit mass, g = F/m	N kg^{-1}
Electric	charge	force per unit positive charge, E = F/Q	N C^{-1} (= V m^{-1})
Magnetic	moving charge or current	B from F = BIl	T

MARK-SCHEME WORDING

- Definition: "region / area where a **charged particle** experiences a **force**". "Region where a charge feels a force" is enough; "region of force" alone is not.
- Sign of a charge from a diagram: "**negative**, because the field direction is the direction of the force on a **positive** charge" (e.g. Earth's surface, where the fair-weather field points downwards).

Often asked as: one-mark definition; MCQ reading the sign of charges X and Y from arrowed field lines.

COMMON MISTAKE

Reading an arrow as the direction an **electron** (or any negative charge) moves. Field lines show the force on a **positive** charge.

SPEC 93 Electric field strength: E = F/Q

WHAT THIS SPEC POINT ASKS

Electric field strength is defined as E = F/Q; use this equation.

Electric field strength

FORMULA SHEET

$$E = \frac{F}{Q}$$

E: electric field strength / N C^{-1} F: force on the charge / N Q: charge / C

- E = electric field strength (N C^{-1}), F = force on the charge (N), Q = charge (C).
- In words: the **force per unit positive charge** at that point.
- E is a **vector**. Its direction is the direction of the force on a positive charge.
- Rearranged, **F = EQ** gives the force on any charge in a known field. For a negative charge, F points opposite to E.
- Once you have F, use the rest of mechanics: a = F/m, work done = F × distance, or combine with the weight.

WORKED EXAMPLE

force and acceleration of a proton

A proton sits in a uniform field of 5.0 × 10^{3} N C^{-1}.

F = EQ = 5.0 × 10^{3} × 1.60 × 10^{-19} = **8.0 × 10^{-16} N**, in the direction of the field.

a = F/m = 8.0 × 10^{-16} ÷ 1.67 × 10^{-27} = **4.8 × 10^{11} m s^{-2}**

An electron in the same field feels the same size of force, but in the **opposite** direction, and its acceleration is about 1800 times larger.

WORKED EXAMPLE

An electron is in a uniform electric field of strength $5.0 \times 10^4\ \text{N C}^{-1}$. Calculate the force on it and its acceleration, and compare the force with its weight.

Show solution

$$F = EQ = 5.0 \times 10^4 \times 1.60 \times 10^{-19} = 8.0 \times 10^{-15}\ \text{N}$$

$$a = \frac{F}{m} = \frac{8.0 \times 10^{-15}}{9.11 \times 10^{-31}} = 8.8 \times 10^{15}\ \text{m s}^{-2}$$

The weight is $mg = 9.11 \times 10^{-31} \times 9.81 = 8.9 \times 10^{-30}\ \text{N}$, about 10^{15} times smaller than the electric force. That's why gravity is ignored for electrons and ions in electric fields. The force acts in the opposite direction to the field, because the electron is negative.

MARK-SCHEME WORDING

- "Use of F = EQ", then "F − W to find the resultant force" if the particle also has weight, then "use of F = ma".
- Field strength units: N C⁻¹ **or** V m⁻¹ are both accepted.

Often asked as: force on a charged drop or ion, then its acceleration; force between plates, then the work done over a distance.

COMMON MISTAKE

Using V/d as if it were the force. V/d is the **field**; multiply by Q to get the force.

Using the total of two charges for Q when only one charge is in the field.

SPEC 94 Coulomb's law: the force between two charges

WHAT THIS SPEC POINT ASKS

Use $F = Q_1Q_2/4\pi\varepsilon_0 r^2$ for the force between two point charges.

Coulomb's law

FORMULA SHEET

$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$$

Q_1, Q_2: the two charges / C r: distance between their centres / m
ε_0: permittivity of free space, $8.85 \times 10^{-12}\ \text{F m}^{-1}$

- F = force between the charges (N), Q_1 and Q_2 = the charges (C), r = separation of their **centres** (m).
- ε_0 = permittivity of free space = 8.85×10^{-12} F m^{-1} (data booklet). The constant $1/4\pi\varepsilon_0 = 8.99 \times 10^9$ N m^2 C^{-2} is accepted too: $F = kQ_1Q_2/r^2$.
- It is an **inverse-square law**: double r and F falls to **a quarter**; triple r and F falls to a ninth.
- Charged **spheres** behave as point charges at their centres.
- Sign: a positive answer means repulsion (like charges); a negative one means attraction. Most questions only want the size.
- The same-sized force acts on **each** charge, in opposite directions (Newton's third law).

WORKED EXAMPLE

two small charged spheres

Charges of +3.0 nC and +5.0 nC have centres 2.5 cm apart.

F = (8.99 × 10⁹ × 3.0 × 10⁻⁹ × 5.0 × 10⁻⁹) ÷ (0.025)² = 1.35 × 10⁻⁷ ÷ 6.25 × 10⁻⁴ = **2.2 × 10⁻⁴ N**, repulsive.

Convert first: nC → × 10⁻⁹ C; cm → m.

WORKED EXAMPLE

the hydrogen atom

An electron and a proton are 5.3 × 10⁻¹¹ m apart.

F = 8.99 × 10⁹ × (1.60 × 10⁻¹⁹)² ÷ (5.3 × 10⁻¹¹)² = **8.2 × 10⁻⁸ N** (attractive)

If this provides the centripetal force: $mv^2/r = F \rightarrow v = \sqrt{(Fr/m)} = \sqrt{(8.2 \times 10^{-8} \times 5.3 \times 10^{-11} \div 9.11 \times 10^{-31})}$ = **2.2 × 10⁶ m s⁻¹**

WORKED EXAMPLE

In a simple model of a hydrogen atom, the electron orbits the proton at a distance of 5.3×10^{-11} m. (a) Calculate the electric force between them. (b) Compare it with the gravitational force, 3.6×10^{-47} N. (c)

The electric force provides the centripetal force. Find the electron's speed.

Show solution

(a) Both charges have size $e = 1.60 \times 10^{-19}$ C:

$$F = \frac{8.99 \times 10^{9} \times (1.60 \times 10^{-19})^{2}}{(5.3 \times 10^{-11})^{2}} = \frac{2.30 \times 10^{-28}}{2.81 \times 10^{-21}} = 8.2 \times 10^{-8}\ \text{N}$$

The charges are opposite, so the force is attractive.

(b) $8.2 \times 10^{-8} / 3.6 \times 10^{-47} \approx 2 \times 10^{39}$. Gravity is completely negligible inside atoms.

(c) Using circular motion, $F = mv^{2}/r$:

$$v = \sqrt{\frac{Fr}{m}} = \sqrt{\frac{8.2 \times 10^{-8} \times 5.3 \times 10^{-11}}{9.11 \times 10^{-31}}} = 2.2 \times 10^{6}\ \text{m s}^{-1}$$

That is under 1% of the speed of light, so the non-relativistic formulas are fine here.

MARK-SCHEME WORDING

- "Use of $F = Q_1Q_2/4\pi\varepsilon_0 r^2$ (accept $F = kQ_1Q_2/r^2$)". The sign of F is optional.
- For nuclei, a separate mark for the charge: alpha = **2e**, gold nucleus = **79e**, each e = 1.60×10^{-19} C.

Often asked as: force between spheres or particles; separation for a given force; MCQ on F when r doubles (F/4); force on an electron in an atom set equal to mv^2/r.

COMMON MISTAKE

Not squaring r, or squaring the charges (the classic MCQ distractors).

Using the diameter for r, or measuring from the surfaces instead of the centres.

Forgetting the unit on the final answer.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Charged spheres on threads:** the thread makes angle θ with the vertical; resolving gives **tan θ = F/W**, so F = mg tan θ, and then Q from Coulomb's law (WPH14 Jan 2022, Jun 2023).

- **Coulomb force then motion:** F = ma, then a time or distance with suvat, compared with a claim (WPH14 Jun 2024).
- **Atom as circular motion:** Coulomb force = mv²/r for the electron (WPH14 Jan 2026).

SPEC 95 Field strength of a point charge

WHAT THIS SPEC POINT ASKS

Use E = Q/4πε₀r² for the electric field of a point charge.

Field strength of a point charge

FORMULA SHEET

$$E = \frac{Q}{4\pi\varepsilon_0 r^2}$$

Q: charge making the field / C r: distance from the charge / m

E around a point charge

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Place a small positive test charge q a distance r from a point charge Q. Coulomb's law gives the force on it:

$$F = \frac{Qq}{4\pi\varepsilon_0 r^2}$$

2. Field strength is force per unit charge, so divide by q:

$$E = \frac{F}{q} = \frac{Qq}{4\pi\varepsilon_0 r^2\, q} \quad \Rightarrow \quad \boxed{E = \frac{Q}{4\pi\varepsilon_0 r^2}}$$

The test charge cancels: the field belongs to Q alone, whether or not anything is there to feel it.

- E = field strength (N C⁻¹), Q = the charge **making** the field (C), r = distance from its centre (m).
- This comes from E = F/Q with Coulomb's law: put a test charge q at r; F = Qq/4πε₀r², so F/q = Q/4πε₀r².
- **Radial** field: lines spread out from the charge, so E ∝ 1/r². Double the distance, quarter the field.

- Direction: **away** from a positive charge, **towards** a negative charge.
- A charged conducting **sphere** gives the same field outside it as a point charge at its centre.

More than one charge

Fields are **vectors**. Work out each field separately (size from the formula, direction from the sign of the charge), then add them as vectors. On a straight line: same direction → **add**; opposite directions → **subtract**.

WORKED EXAMPLE

field between two charges

+6.0 nC and −4.0 nC are 0.20 m apart. Find E at the midpoint (0.10 m from each).

From +6.0 nC: $E_1 = 8.99 \times 10^9 \times 6.0 \times 10^{-9} \div 0.10^2 = 5390$ N C^{-1}, pointing **away** from it (towards the −4.0 nC).

From −4.0 nC: $E_2 = 8.99 \times 10^9 \times 4.0 \times 10^{-9} \div 0.10^2 = 3600$ N C^{-1}, pointing **towards** it (also towards the −4.0 nC).

Same direction, so add: E = 5390 + 3600 = **9.0×10^3 N C^{-1}** towards the negative charge.

With two **like** charges the fields at the midpoint point opposite ways and you subtract. Equal like charges give **zero** field at the midpoint: a **neutral point**.

WORKED EXAMPLE

The dome of a Van de Graaff generator is a sphere of radius 0.15 m carrying a charge of $+2.0\ \mu\text{C}$. Calculate the field strength at its surface and 0.30 m from its centre.

Show solution

$$E_{0.15} = \frac{8.99 \times 10^9 \times 2.0 \times 10^{-6}}{0.15^2} = 8.0 \times 10^5 \text{ N C}^{-1}$$

$$E_{0.30} = \frac{8.99 \times 10^9 \times 2.0 \times 10^{-6}}{0.30^2} = 2.0 \times 10^5 \text{ N C}^{-1}$$

Doubling the distance from the centre quarters the field. Notice that 0.30 m from the centre is only 0.15 m from the surface: always measure from the centre.

MARK-SCHEME WORDING

- "Adds E due to X to E due to Y" **with the directions considered**.
- Units N C^{-1} or V m^{-1}.

Often asked as: field at a point from one or two charges; the charge on the Earth from the field at its surface; field in a molecule.

COMMON MISTAKE

Subtracting when both fields point the same way (nearly half of one cohort did this).

Using Q_1Q_2 in a field question. A field needs only the **source** charge.

Giving field strength in V instead of V m^{-1}.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Surface charge density** = Q/A with A = $4\pi r^2$ for a sphere (WPH14 Jan 2020).
- **Capacitance of an isolated sphere**, C = $4\pi\varepsilon_0 r$, used with Q = CV to find the charge on a sphere (WPH14 Oct 2023).

SPEC 96 Electric field and electric potential

WHAT THIS SPEC POINT ASKS

Know and understand how electric field strength is related to electric potential.

Electric potential

- The **electric potential** V at a point is the **work done per unit positive charge** in bringing a small positive charge from **infinity** to that point. Unit: volt, V = J C^{-1}.
- Potential is a **scalar**: it has no direction, only a sign.
- Potential at infinity is taken as **zero**. Near a positive charge V is positive; near a negative charge V is negative.
- **Potential difference** between two points is the work done per unit charge moving between them. Moving a charge Q through a p.d. ΔV transfers energy:

Energy transferred by a p.d.

$$\Delta W = Q\,\Delta V$$

ΔW: work done / J Q: charge / C ΔV: potential difference / V

Field = potential gradient

- The field strength is the **rate of change of potential with distance**, with a minus sign:

Field strength and potential gradient

$$E = -\frac{dV}{dr}$$

E: electric field strength / V m^{-1} V: electric potential / V r: distance / m

in a uniform field, $E = -\Delta V / \Delta x$

Field strength is minus the potential gradient

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Move a small positive charge q a short distance Δr in the direction of the field E. The field pushes it with force qE, so the field does work on it:

$$W = qE\,\Delta r$$

2. That energy comes out of the charge's electric potential energy, which falls by $qE\,\Delta r$. Per unit charge, the potential falls by $E\,\Delta r$:

$$\Delta V = -E\,\Delta r$$

3 Rearrange:

$$\boxed{E = -\frac{\Delta V}{\Delta r}}$$

The field strength is the gradient of the V–r graph with its sign reversed. The minus sign means the field points "downhill", from high potential to low potential.

- The minus sign means the field points from **high potential to low potential** (downhill).
- On a graph of V against r: field strength = **minus the gradient**. A steeper graph means a stronger field.
- On a graph of E against r: the **area** under the graph between two distances = the potential difference between them.
- Units check: V m^{-1} = (J C^{-1}) m^{-1} = (N m C^{-1}) m^{-1} = **N C^{-1}**. The two units of E are the same thing.

WORKED EXAMPLE

energy and field from potentials

An alpha particle (charge 2e) moves from the 40 V equipotential to the 10 V equipotential.

ΔW = QΔV = 2 × 1.60 × 10^{-19} × 30 = **9.6 × 10^{-18} J** (= 60 eV) of kinetic energy gained.

If those equipotentials were 3.0 mm apart in a uniform field: E = ΔV/Δx = 30 ÷ 3.0 × 10^{-3} = **1.0 × 10^{4} V m^{-1}**.

WORKED EXAMPLE

(a) Calculate the potential at the surface of the Van de Graaff dome from the previous example (radius 0.15 m, charge $+2.0\ \mu\text{C}$) and 0.30 m from its centre. (b) A proton moves from a point where the potential is 60 kV to a point where it is 20 kV. Starting from rest, how fast is it moving?

Show solution

(a)

$$V_{0.15} = \frac{8.99 \times 10^{9} \times 2.0 \times 10^{-6}}{0.15} = 1.2 \times 10^{5}\ \text{V}, \qquad V_{0.30} = 6.0 \times 10^{4}\ \text{V}$$

Doubling the distance halved the potential but quartered the field. For a point charge, $E = V/r$: $1.2 \times 10^{5}/0.15 = 8.0 \times 10^{5}\ \text{N C}^{-1}$, matching the earlier answer.

(b) The proton falls through a potential difference of $60 - 20 = 40\ \text{kV}$. The work done on it by the field becomes kinetic energy:

$$E_k = Q\,\Delta V = 1.60 \times 10^{-19} \times 4.0 \times 10^{4} = 6.4 \times 10^{-15}\ \text{J}$$

$$v = \sqrt{\frac{2E_k}{m}} = \sqrt{\frac{2 \times 6.4 \times 10^{-15}}{1.67 \times 10^{-27}}} = 2.8 \times 10^{6}\ \text{m s}^{-1}$$

MARK-SCHEME WORDING

- Field strength is the "potential gradient"; "potential is the area under the E–r graph" (MCQ explanations).
- Electric potential is a **scalar** (MCQs listing scalars and vectors).
- Showing V ∝ 1/r from a diagram: measure r to two equipotentials and show "V × r is constant".

Often asked as: MCQ on area under E–r or gradient of V–r; energy in eV gained between equipotentials.

EXAM TIP

In eV, the energy gained is just (charge in units of e) × (p.d. in volts). An alpha crossing 30 V gains 2 × 30 = 60 eV. Convert to J only at the end.

SPEC 97 Uniform field between parallel plates: E = V/d

WHAT THIS SPEC POINT ASKS

Use E = V/d for the uniform field between parallel plates.

Uniform field between parallel plates

FORMULA SHEET

$$E = \frac{V}{d}$$

V: p.d. between the plates / V d: plate separation / m

E = V/d for parallel plates

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Move a positive charge Q from the negative plate to the positive plate, a distance d, against the field. The force needed is $F = EQ$, which is constant because the field is uniform. The work done is

$$W = Fd = EQd$$

2. By the definition of potential difference, moving charge Q through p.d. V takes work

$$W = QV$$

3. The two must be equal. Q cancels:

$$EQd = QV \quad \Rightarrow \quad \boxed{E = \frac{V}{d}}$$

So the unit of field strength can also be written V m⁻¹. This is just $E = -\Delta V / \Delta r$ for a field whose potential falls steadily from one plate to the other.

- E = field strength (V m⁻¹), V = p.d. between the plates (V), d = plate separation (m).
- Why: moving a charge Q from one plate to the other takes work QV = Fd, so F/Q = V/d.
- Between the plates the field is **uniform**: the same size and direction everywhere (except near the edges). The force on a charge is the same wherever it is between the plates.
- The field points from the **positive** plate to the **negative** (or earthed, 0 V) plate.

A charge entering the field sideways

An electron entering between horizontal plates at speed u along the plates behaves like a **projectile**:

- **Along** the plates: no force, so **constant speed**: t = length ÷ u.
- **Across** the plates: constant force F = eE, so constant acceleration a = eE/m. Use s = ½at² for the sideways deflection.
- The path between the plates is a **parabola**; after leaving the plates it is a straight line.

WORKED EXAMPLE

deflection of an electron beam

Plates 2.5 cm apart have a p.d. of 500 V and are 5.0 cm long. An electron enters midway between them at 2.0×10^7 m s⁻¹, parallel to the plates.

E = V/d = 500 ÷ 0.025 = 2.0×10^4 V m⁻¹

a = eE/m = $(1.60 \times 10^{-19} \times 2.0 \times 10^4) \div 9.11 \times 10^{-31} = 3.51 \times 10^{15}$ m s⁻²

$t = 0.050 \div 2.0 \times 10^{7} = 2.5 \times 10^{-9}$ s

$s = \tfrac{1}{2}at^2 = 0.5 \times 3.51 \times 10^{15} \times (2.5 \times 10^{-9})^2 = 0.011$ m = **1.1 cm** towards the positive plate.

That is less than the 1.25 cm to the plate, so the electron gets through.

WORKED EXAMPLE

An electron travelling at $3.0 \times 10^{7}\ \text{m s}^{-1}$ enters the space midway between two parallel plates 8.0 cm long and 2.0 cm apart, moving parallel to them. The p.d. between the plates is 150 V. Calculate (a) the field strength, (b) the acceleration of the electron, (c) its deflection as it leaves the plates and (d) the angle its path then makes with its original direction.

Show solution

(a)

$$E = \frac{V}{d} = \frac{150}{0.020} = 7500\ \text{V m}^{-1}$$

(b)

$$a = \frac{eE}{m} = \frac{1.60 \times 10^{-19} \times 7500}{9.11 \times 10^{-31}} = 1.32 \times 10^{15}\ \text{m s}^{-2}$$

(c) Horizontally the speed is constant: $t = L/v_0 = 0.080/3.0 \times 10^{7} = 2.67 \times 10^{-9}$ s. Vertically it starts from rest:

$$y = \tfrac{1}{2}at^2 = \tfrac{1}{2} \times 1.32 \times 10^{15} \times (2.67 \times 10^{-9})^2 = 4.7 \times 10^{-3}\ \text{m} = 4.7\ \text{mm}$$

That is less than the 10 mm gap to the plate, so the electron gets through.

(d) Vertical velocity at the exit: $v_y = at = 1.32 \times 10^{15} \times 2.67 \times 10^{-9} = 3.5 \times 10^{6}\ \text{m s}^{-1}$.

$$\theta = \tan^{-1}\left(\frac{3.5 \times 10^{6}}{3.0 \times 10^{7}}\right) = 6.7^{\circ}$$

MARK-SCHEME WORDING

- Projectile wording: "resultant force on the electrons in the **vertical** direction, so they **accelerate vertically**; in the **horizontal** direction they have **constant speed**".
- Uniform-field diagrams: "at least three **equally spaced parallel** lines", "touching both plates", "arrow from + to −".
- Charge as a multiple of e: "Q = 5e" or "integer multiple of 1.6×10^{-19} C". A value like 4.97e should be recognised as **5e**.

Often asked as: force on a particle between plates; speed gained crossing the gap (Ek = QV); deflection of an electron beam; Millikan-type drops; charged spheres shuttling between plates.

COMMON MISTAKE

Leaving d in cm or mm.

Using V/d as the force instead of the field.

Applying suvat along the plates as if there were an acceleration that way.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Millikan's oil-drop method:** a drop is held still when the electric force balances its weight: **QV/d = mg**. Example: a drop of mass 2.0×10^{-15} kg is held between plates 1.2 cm apart by 490 V. $Q = mgd/V = 2.0 \times 10^{-15} \times 9.81 \times 0.012 \div 490 = 4.8 \times 10^{-19}$ C = **3e**. All charges come out as whole-number multiples of e, so charge is **quantised** (WPH14 Jan 2021, Jan 2025).
- The drop's radius comes from its terminal velocity with Stokes' law, $6\pi\eta rv = (4/3)\pi r^3\rho g$ (WPH14 Jan 2021).
- A charged sphere touching a plate takes that plate's sign, is repelled to the other plate, swaps sign and returns: it carries charge across and a current flows (WPH14 Oct 2023).

SPEC 98 Potential in a radial field

WHAT THIS SPEC POINT ASKS

Use $V = Q/4\pi\varepsilon_0 r$ for the potential near a point charge.

Potential in a radial field

FORMULA SHEET

$$V = \frac{Q}{4\pi\varepsilon_0 r}$$

V: electric potential / V Q: charge / C r: distance from the charge / m

- V = potential (V), Q = the charge making the field (C), r = distance from its centre (m).
- **V ∝ 1/r** (but E ∝ 1/r²). Triple the distance and the potential falls to a third.

- Sign follows Q: positive charges make positive potentials, negative charges negative potentials.
- **Electric potential energy** of a second charge q at distance r: $E_p = qV = Qq/4\pi\varepsilon_0 r$.
- **Several charges:** potentials are scalars, so just **add them with their signs**.

WORKED EXAMPLE

potential and work done

A +4.0 nC point charge. Potential at 0.30 m: $V = 8.99 \times 10^9 \times 4.0 \times 10^{-9} \div 0.30 =$ **120 V**.

At 0.10 m: V = 360 V. Work done moving a +2.0 nC charge from 0.30 m to 0.10 m:

$W = q\Delta V = 2.0 \times 10^{-9} \times (360 - 120) =$ **4.8×10^{-7} J**

Closest approach of an alpha particle

An alpha particle fired straight at a nucleus slows down as it climbs the potential "hill". At the closest approach it is momentarily at rest, so **all its kinetic energy has become electrical potential energy**:

Closest approach of an alpha particle

$$E_k = \frac{Q_\alpha Q_N}{4\pi\varepsilon_0 r_{\min}}$$

E_k: initial kinetic energy / J Q_α, Q_N: charges of the alpha particle and the nucleus / C
$r_{\min}$: distance of closest approach / m

WORKED EXAMPLE

how close does a 5.0 MeV alpha get to a gold nucleus?

$E_k = 5.0 \times 10^6 \times 1.60 \times 10^{-19} = 8.0 \times 10^{-13}$ J. Charges: alpha = 2e, gold = 79e.

$r = (8.99 \times 10^9 \times 2 \times 79 \times (1.60 \times 10^{-19})^2) \div 8.0 \times 10^{-13} =$ **4.5×10^{-14} m**

This is an **upper limit** for the radius of the gold nucleus: the alpha never touched it.

MARK-SCHEME WORDING

- "Recognises Q is 2 (× unit charge)" for an alpha: a separate mark.
- Why the formula works for a nucleus: "the field is **radial**", "the nucleus can be treated as a **point charge**", "no other charges nearby", "distance measured from the **centre**".
- Two charges: "potential at each point is the **sum** of the potential due to X and the potential due to Y".

- Alpha not aimed head-on: it still has speed at its nearest point, so "it is not as close to the nucleus" and the maximum force is smaller.

Often asked as: potential or potential energy at a distance; work done moving a charge between radii; closest approach from an energy in MeV; p.d. between two points near two charges.

COMMON MISTAKE

Forgetting that gold is 79e and alpha is 2e (or using charge 1).

Neglecting the "M" in MeV.

Using r^2 (that is the field formula).

Adding potentials as vectors: they are scalars.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Hydrogen atom energy:** the potential energy of the electron is negative, $-e^2/4\pi\varepsilon_0 r = -4.3 \times 10^{-18}$ J at 5.3×10^{-11} m (WPH14 Jan 2026).
- **Maximum acceleration** at closest approach: F from Coulomb's law at r_{min}, then a = F/m (WPH14 Jan 2021).

SPEC 99 Field lines and equipotentials

WHAT THIS SPEC POINT ASKS

Draw and interpret field-line and equipotential diagrams for radial and uniform electric fields.

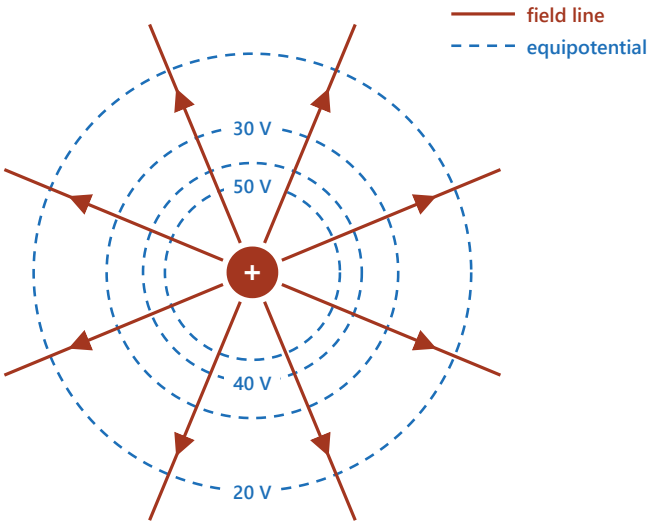
Rules for field lines

- A field line shows the direction of the force on a **positive** charge: draw an **arrow** on every line.
- Lines start on positive charges and end on negative charges (or go off to infinity).
- **Closer lines = stronger field**. Equally spaced parallel lines = **uniform** field.
- Lines **never cross** (the field can only have one direction at a point).
- Lines meet a conducting surface (a plate or sphere) at **90°**.

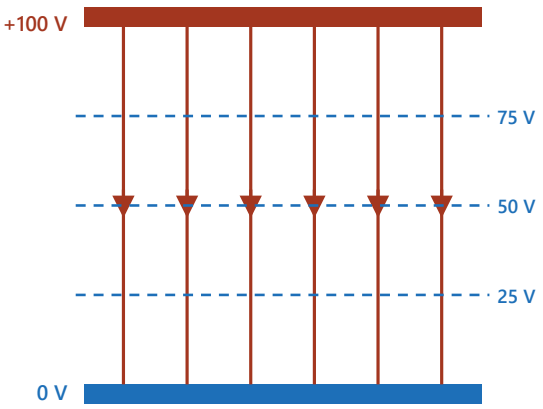
Rules for equipotentials

- An **equipotential** is a line (or surface in 3-D) joining points at the **same potential**.

- Equipotentials are always **perpendicular to field lines**.
- **No work** is done moving a charge along an equipotential ($\Delta V = 0$, so $Q\Delta V = 0$).
- Draw them at **equal steps in potential** (e.g. every 10 V). Where they are **closer** together the field is **stronger** ($E = -\Delta V/\Delta x$).
- The surface of a conductor is an equipotential.



Radial field: equipotentials spread out



Uniform field: equally spaced lines and equipotentials

Left: a positive point charge. Field lines are radial with arrows outwards; equipotentials (here every 10 V) are circles that get further apart. Right: parallel plates. Field lines are parallel, equally spaced and touch both plates; equipotentials are equally spaced lines parallel to the plates.

Radial field (point charge or charged sphere)

- Field lines: straight, **radial**, equally spaced in angle, arrows **out** of a positive charge and **in** to a negative one. They spread out, showing the field weakening.
- Equipotentials: **concentric circles**. Because $V \propto 1/r$, equal steps in V give circles that are **further apart** as r increases.

V	120 V	90 V	60 V	30 V
r (for V × r = 36 V m)	0.30 m	0.40 m	0.60 m	1.20 m
gap to next	0.10 m	0.20 m	0.60 m	

Uniform field (parallel plates)

- Field lines: **parallel, equally spaced**, straight, from the + plate to the − plate, touching both. (Ignore slight bulging at the edges.)
- Equipotentials: **parallel to the plates**, **equally spaced**. With 100 V across, the 50 V line is halfway.

Two charges (sketches you should recognise)

- **Opposite charges:** lines curve from + to −, crowded between them. The 0 V equipotential is the line halfway between equal and opposite charges.
- **Like charges:** lines from each charge bend away from the other; there is a **neutral point** (E = 0) between them. For unequal charges the neutral point is **nearer the smaller** charge.

MARK-SCHEME WORDING

- Radial: "at least **four** radial straight lines", "**equally spaced**", "arrow(s) **outward**" (or inward for negative).
- Uniform: "at least three **equi-spaced** parallel lines", "**touching the plates**", "arrow". Curved lines at the edges are ignored.
- Equipotentials round a point charge: "circles centred on the charge" and "**spacing increasing** with distance".
- Two like charges: "neutral point equidistant", "lines diverging", "lines closer between the charges than outside".

Often asked as: draw the field round a charged sphere or between plates; add equipotentials; read the sign of charges from arrows; "is the field line the path the charge follows?"

COMMON MISTAKE

Drawing radial equipotentials **equally spaced**. They must spread out, because $V \propto 1/r$.

Missing arrows or arrows the wrong way. Use a ruler.

Lines that stop short of the plates, or that cross.

EXAM TIP

A field line is **not** the path of a charge in general. A charge released at rest on a **straight** field line does move along it, but on a curved line its velocity cannot follow the changing direction of the force.

Quick checks

QUICK CHECK

The distance between two point charges is doubled. What happens to the force between them?

- It halves.
- It falls to a quarter.
- It doubles.
- It falls to an eighth.

$F \propto 1/r^2$, so doubling r gives $1/2^2 = 1/4$ of the force. "Halves" would be the answer for the potential, which goes as $1/r$.

QUICK CHECK

Which of these is a unit of electric field strength?

- V m^{-1}
- V m
- N C
- J C^{-1}

From $E = V/d$, field strength is in V m^{-1}, which is the same as N C^{-1}. J C^{-1} is the volt, the unit of potential.

QUICK CHECK

The electric field at a point points due east. Which way is the force on an electron placed there?

- East
- West
- There is no force on an electron at rest.
- At right angles to the field

The field direction is the direction of the force on a **positive** charge. The electron is negative, so it is pushed the opposite way. Electric forces act on charges whether or not they are moving.

QUICK CHECK

Equipotentials around a point charge are drawn at equal steps of potential. Moving away from the charge, they

- are equally spaced.
- get further apart.
- get closer together.
- are straight, parallel lines.

The field gets weaker further out, so the potential changes more slowly with distance: you have to go further to drop by the same number of volts.

QUICK CHECK

The p.d. across a pair of parallel plates is kept the same while their separation is doubled. What happens to the force on a charge between them?

- It doubles.
- It halves.
- It stays the same.
- It falls to a quarter.

$E = V/d$, so doubling d halves E, and $F = EQ$ halves too. It is not an inverse-square effect: the field between plates is uniform.

QUICK CHECK

Point P is midway between a charge $+Q$ and a charge $-Q$. Which statement is correct at P?

- $V = 0$ and $E = 0$
- $V = 0$ but E is not zero; it points towards $-Q$.
- $E = 0$ but V is not zero.
- V and E both point towards $+Q$.

Potential is a scalar: $+kQ/r$ and $-kQ/r$ add to zero. Field is a vector: both charges' fields at P point the same way (away from $+Q$, towards $-Q$), so they add rather than cancel. Potential has no direction at all.